GEOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Geosource Energy 1508 Hwy 54, Caledonia Ontario N3W 2G9 ATTN: Peter Ndep, MSc, PMP, CEM, P.Eng	2023-08-15	QU-0108 **Reference** CA ON Geosource Glanbrook NPH

Understanding of project

The proposed Glanbrook NP Housing Corporation is considering the installation of a ground source heat pump (GSHP) system to provide heating, cooling, and possible domestic hot water for the facility. The project includes 69 one-bedroom and 36 two-bedroom units and a community building space within an area of approximately 9,287 m^2 (99,928 ft^2). The building is situated on a level site of approximately 30,300 m^2 (326,000 ft^2).

Scope of work

- Develop a detailed 8,760 hourly energy model in Trane Trace based on drawings and information provided by the client,
- Estimate monthly and annual energy consumption profiles,
- Review of thermal conductivity test performed by others,
- GHX modelling using GLD2016 software for GHX size and location,
- Configuration of GHX circuit piping for the recommended GHX configuration,
- Detailed design of fusion welded headers to connect GHX circuits to supply / return runout piping,
- Supply / return runout piping from GHX headers to manifold in geothermal vault or building,
- Written specifications for GHX piping for circuits, headers, and supply / return runout piping,
- Specifications for heat transfer fluid and estimated heat transfer fluid volume,
- GHX manifold design up to isolation valves or heat exchanger between heat pump system piping and GHX,
- Pressure-drop calculations for the GHX and recommended circulation pump sizing and control,
- Detailed AutoCAD drawings of proposed GHX, including layout on building site plan, header design, borehole design and supply / return runout pairs,
- Coordination with mechanical engineer in development of mechanical system schematic for interior mechanical system,
- Development of QA / QC program for installation of GHX,

- Pre-construction meeting with Geosource Energy, project site supervisor, mechanical contractor, and other stakeholders in the project to avoid duplication and ensure coordination of installation,
- Develop detailed operating and maintenance manual for GHX.

Deliverables

- 8760 HVAC loads and energy consumption in a spreadsheet,
- Design Documents,
- Preliminary Geothermal Design,
- Detailed Geothermal Design – Issue for Permit (signed and sealed),
- Detailed Geothermal Design – Issue for Tender (signed and sealed),
- Detailed Geothermal Design – Issue for Construction (signed and sealed),
- 1 site visit,
- Post Construction, Commissioning, and System Signoff (signed and sealed).

Pricing

Description	Rate		Qty	Line Total
Detailed Design	$55,500.00 +ON HST		1	$55,500.00 $7,215.00
Total (CAD)				**$62,715.00**

Milestones

1) Preliminary Geothermal Design
    a. Estimated time for completion: 5 weeks
    b. Invoicing amount: $19,260.00

2) Detailed Geothermal Design – Issue for Permit
    a. Estimated time for completion: 3.5 weeks
    b. Invoicing amount: $9,600.00

3) Detailed Geothermal Design – Issue for Tender
    a. Estimated time for completion: 2.5 weeks
    b. Invoicing amount: $8,880.00

4) Detailed Geothermal Design – Issue for Construction
    a. Estimated time for completion: 2.5 weeks
    b. Invoicing amount: $8,880.00

5) Post Construction, Commissioning & System Signoff
    a. Estimated time for completion: 2 weeks
    b. Invoicing amount: $8,880.00

Terms

- Additional work out of scope for any phase will be charged at $250/hr.
- Invoices will be submitted once each milestone is completed.
- Payment is expected within 60 days post receipt of invoice.
- Timeline dependent on information receival.
- Pricing includes 1 site visit for commissioning and system signoff.
    - Geosource to provide GEOptimize with photos and information for installation QA/QC protocols.

Name:

Title:

Signature: